INCH-POUND

MIL-C-24640/12C
26 June 1995
SUPERSEDING
MIL-C-24640/12B
31 July 1989

MILITARY SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT WEIGHT, MULTI PAIR AWG 26, TYPE 2XO

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.

CONSTRUCTION
(Non-Watertight)

FIRST - 19 strand copper conductor in accordance with ASTM B-286, tinned strands in accordance with ASTM B-33. AWG size 26 maximum diameter 0.022 inches.

* SECOND - Component wires shall be MIL-W-81044/12 and shall be tested in accordance with the component wire ratings and requirements listed in this slash sheet. Quality conformance testing shall be in accordance with Table VI of MIL-C-24640.. Twisted pair identification code, one black and one white, applied using method 6 or telephone color code applied using method 3.

THIRD - Two conductors twisted together. Pair lay length shall be in accordance with Tables II and III.

FOURTH - Cable assembly shall be in accordance with Table III. Cabling sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used as necessary, to form a round cable.

FIFTH - Binder tape applied helically with overlap.

SIXTH - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface Transfer Impedance requirements as listed herein.

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CONSTRUCTION (continued):

SEVENTH - Crosslinked polyolefin jacket. (See Table I for thickness).

TABLE I. Construction details.

PIN M24640/12	Type number	Number of component	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum	Cable diameter (inch) min	max	Cable weight (pounds per 1000 feet) (max)
-01UO	2XO-6	6	0.220	0.038	0.305	0.329	81.6
-02UO	2XO-18	18	.356	.041	.417	.449	140.0
-03UO	2XO-24	24	.420	.044	.473	.509	175.0
-04UO	2XO-42	42	.536	.044	.565	.609	275.0
-05UO	2XO-60	60	.622	.047	.641	.691	334.0
-06UO	2XO-77	77	.714	.047	.728	.785	395.0

TABLE II. Pair lay[1/]

Pair no.	Pair lay (inches)	Pair no.	Pair lay (inches)	Pair no.	Pair lay (inches)
1	1.5	27	1.0	53	2.5
2	1.0	28	2.0	54	3.5
3	2.0	29	3.0	55	4.5
4	3.0	30	4.5	56	5.5
5	4.5	31	1.0	57	1.0
6	1.0	32	2.0	58	1.5
7	2.0	33	3.0	59	2.0
8	3.0	34	4.5	60	2.5
9	1.5	35	1.0	61	3.0
10	2.5	36	2.0	62	4.5
11	5.5	37	3.0	63	5.5
12	1.5	38	4.5	64	1.0
13	2.5	39	1.0	65	1.5
14	3.7	40	2.0	66	2.0
15	5.5	41	3.0	67	2.5
16	1.5	42	4.5	68	3.0
17	2.5	43	1.0	69	4.5
18	3.7	44	1.5	70	5.5
19	1.5	45	2.0	71	1.0
20	2.5	46	2.5	72	1.5
21	3.7	47	3.0	73	2.0
22	5.5	48	4.5	74	2.5
23	1.0	49	5.5	75	3.0
24	2.0	50	1.0	76	4.5
25	3.0	51	1.5	77	5.5
26	4.5	52	2.0		

[1/] All lays are left handed.

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TABLE III. Assembly

Type number	Core	Layer 1	Assembly Layer 2	Layer 3	Layer 4
2XO-6	- - -	Pairs 1-6			
2XO-18	Pair 1	Pairs 2-6	Pairs 7-18		
2XO-24	Pairs 1-2	Pairs 3-10	Pairs 11-24		
2XO-42	- - -	Pairs 1-8	Pairs 9-22	Pairs 23-42	
2XO-60	Pair 1	Pairs 2-7	Pairs 8-19	Pairs 20-36	Pairs 37-60
2XO-77	Pairs 1-3	Pairs 4-12	Pairs 13-27	Pairs 28-49	Pairs 50-77

REQUIREMENTS:

Qualification required.
Construction details: See tables I, II and III.

Component wire:

For the component wire the following additional tests shall be applicable in accordance with MIL-W-81044:

WIRE, RATINGS AND ADDITIONAL REQUIREMENTS:

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 300± 2°C (572± 3.6°F) for 6 hours.
Blocking: 150± 2°C (302± 3.6°F)
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test:
Primary insulation (when tested in lieu of spark test): 6.0 kilovolts (peak) 100 percent test.
Finished wire: 8.0 kilovolts (peak), 3 Khz spark test 5.5KV, 100 percent test.
Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200± 2°C (392± 3.6°F) for 168 hours.
Physical properties of primary insulation:
Tensile strength, 2500 lb/in^2 (min).
Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 300± 2°C (572± 3.6°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz, 100 percent test.
Surface resistance: 500 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150± 2°C (302± 3.6°F).
Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	44.24
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield	1000
Component Shield to shield	Not applicable
Insulation resistance - megohms/1000 feet, minimum	2500
Conductor and shield continuity	No failure
Jacket flaws	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	33
Capacitance unbalance - percent, maximum	15
Characteristic impedance - at 1 MHz, ohms	75 min, 105 max
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	Not applicable
Weight	No failure
Group B:	
Cold bending	No failure
Drip	Not applicable
* Cross link proof test (percent maximum)	
Jacket	50
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Attenuation - at 3 MHz, dB/100 feet, maximum	Not applicable
Gas flame (1 hour)	Not applicable
Group C:	
Physicals (aged)	
Jacket	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
* Tensile strength - percent of unaged, minimum	50
* Elongation - percent of unaged minimum	50
* Permanence of printing (conductor) method 1 only cycles, minimum	25
* Permanence of printing (jacket) - cycles, minimum	125

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	Armor - conformance to material, construction and coverage	Not applicable
	Group C (continued):	
	Shield - conformance to material, construction and coverage	No failure
	Heat distortion - percent of unaged, maximum	30
*	Cable sealant removability	Not applicable
	Shrinkage	No failure
	Mutual inductance - dB/1000 feet, maximum	Not applicable
*	Surface transfer impedance	
	Milliohms per meter, maximum	700
	EMP response, dB, minimum	60
	Pulse response time - ms/500 feet	
	Rise time, maximum	Not applicable
	Decay time, maximum	Not applicable
*	Group D	
	Flame propagation	No Failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Aging and compatibility (cable)	No failure
	Acid gas equivalent - percent, maximum	
	Jacket	2
	Fillers	2
	Insulation	18
	Halogen content - percent, maximum	
	Jacket	0.2
	Fillers	0.2
	Immersion	
	Tensile strength - percent of unaged, minimum	50
	Elongation - percent of unaged minimum	50
*	Smoke index, maximum	
	Jacket	25
	Fillers	45
	Insulation	10
	Toxicity index, maximum	
	Jacket	5
*	Fillers	5
	Insulation	1.5
	Durometer (jacket) - (type A) hardness, minimum	80
	Weathering (jacket)	No failure
	Fungus Resistance	No failure
	Abrasion resistance	No failure
	Gas Flame (3 hour)	Not applicable
	Component wire	No failure

Unit ordering length: 6 - 24 500 feet nominal
42 - 77 250 feet nominal

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Navy - SH

Review Activities:
Army - AV, CR, ER, ME, AR, AL
Navy - EC, CG
DLA - IS

Preparing Activity:
Navy - SH

Agent:
DLA-IS

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